Kotha Sree Janani

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2025	8.57
Narayana Junior College	Class XII (Telangana State Board of Intermediate Education)	2023	97.50%
Greenwood High School	Class X (CBSE)	2021	92.00%

SKILLS

- **Programming Languages:** Python, Java, familiar with C
- **Software, Tools and Libraries:** NumPy, Pandas, Matplotlib, Seaborn, Scikit Learn, TensorFlow/Keras, NLTK, PyTorch, Visual Studio Code
- **Tools and Technologies:** Anaconda, Jupyter Notebook, Google Colab
- **General:** Data Structure and Algorithms
- **Databases:** MySQL

PROJECTS

AI-Driven Defect Recognition and Generative Modeling for Composite Structures Using Microwave NDE:

Under Prof. Thulsiram G, IIT Hyderabad | Aug 2026-Present

- Developed a **CNN from scratch using NumPy** for automated defect detection/localization in 4-channel composite-defect images obtained through Microwave NDE.
- Implemented **convolution, ReLU, max pooling, fully connected layers, forward/backpropagation, and MSE-based coordinate regression**.
- **Future Work:** Develop **Generative AI models using ANN, CNN, and Transformer architectures**, followed by automated defect detection and classification.

Telecom Customer Churn Prediction:

Tech Stack: *Python, Pandas, NumPy, Matplotlib, Seaborn, Scikit-learn, XGBoost*

- Analyzed **7,043 telecom customer records**, performing EDA, correlation analysis, and feature engineering to identify key churn drivers.
- Engineered **interaction features** such as *Tenure* × *Contract Type* and *Monthly Charges* × *Fiber Optic*, followed by Random Forest-based feature selection.
- Built and compared **Logistic Regression, Random Forest, and XGBoost**, achieving **0.84 ROC-AUC, 79% Recall, and 0.62 F1-score** with Random Forest.

Article Authenticity Classifier using NLP:

Tech Stack: Python, Scikit-learn (TF-IDF, Logistic Regression), Pandas, NLTK

- Engineered an end-to-end **NLP pipeline** to clean, preprocess, and structure a large-scale dataset of **44,000+ news articles** using tokenization, lemmatization, stopword removal, and **TF-IDF vectorization**, enabling efficient and reliable text modeling.
- Built and optimized a binary text classification model using Logistic Regression with unigram and bigram TF-IDF features, achieving **~99% accuracy** while maintaining balanced precision and recall on unseen test data.
- Evaluated model performance using **confusion** matrix and **classification** metrics, ensuring reliable detection of fake and authentic news.

EXTRACURRICULAR

- **Volunteer, Learn in Bharat (LIB) 2026 - Next Bharat Ventures:** Contributed to the successful execution and coordination of the program.
- Member of **NSO Football** at IIT Hyderabad
- Secured **2nd place in Football** at **DIESTA 2024**, the inter-departmental sports competition, **IIT Hyderabad**.

RELEVANT COURSES

- **Computer Science & AI:** Introduction to Programming, Introduction to Artificial Intelligence
- **Mathematics:** Calculus I & II ,Elementary Linear Algebra, Differential Equations, Introduction to Probability, Transform Techniques, Intro to Statistics, Complex Variables
- **Additional Courses:** Business Analytics using Python

CERTIFICATIONS

- **Prime AI/ML Course:** Certified from Apna College